

POWERFACTORY

PowerFactory 2024

Technical Reference

Voltage Measurement

StaVmea

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

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https://www.digsilent.de

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March 28, 2024
PowerFactory 2024
Revision 2

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1 General Description

The *Voltage Measurement Device* (*StaVmea*) can be used to measure the voltage at terminals and cubicles. The measured voltage can then for example be fed as a signal into a controller for a RMS or EMT simulation.

The *Voltage Measurement Device* has either to be connected to a measurement point by selecting a terminal or cubicle for the *Measurement Point* ($pbusbar$ parameter) or it can be directly stored at the measurement location - which has to be a cubicle (*StaCubic*). If the cubicle has a circuit-breaker, the measurement device is connected to the circuit-breaker from the side of the edge element.

AC, DC and BI-phase system types can be measured by the *StaVmea* element. With the option *Measurement*, the voltage can be measured at :

- *Single terminal*

  If the model with *Number of phases* =3 is used, PH-E, PH-N and PH-PH quantities can be measured. With this option, not only three-phase, but also two-phase and single-phase terminals can be measured (the non-relevant outputs are then set to 0).

  If the model with *Number of phases* =1 is used, the measured quantities can be selected as PH-E or PH-N quantities (*a, b, c, a-n, b-n, c-n or n*) by selecting the appropriate *Measured Phase*).

- *Between two terminals*

  If the model with *Number of phases* =3 is used, only PH-E quantities can be measured. With this option, not only three-phase, but also two-phase and single-phase terminals can be measured (the non-relevant outputs are then set to 0).

  If the model with *Number of phases* =1 is used, the measured quantities can be selected as PH-E or PH-N quantities (*a, b, c, a-n, b-n, c-n or n*) by selecting the appropriate *Measured Phase*).

  The PH-PH and voltage difference values are calculated as the voltage at the first minus the voltage at the second measurement point.

The outputs of the measurement device can be selected to be phase or transformed quantities (symmetrical components for LF and RMS calculations and $\alpha\beta\gamma$ components for the EMT). depending on the selection of the *Output base* three different options are available:

- *Equivalent to kV values*, where the output value corresponds to the phase value in kV,
- *Based on nominal voltage of connected terminal* where the output is based on the $uknom$ parameter of the connected terminal, and
- *Based on user-defined voltage* where the output has is calculated using the $Unom$ input parameter as base value. If the nominal voltage of the connected/measured terminal is used as $Unom$, this option is equivalent to the second option.

For the EMT calculation, options are also available to measure RMS quantities. This option can be selected on the EMT page and is supported for phase quantities measurements at single terminals.

The voltage magnitudes (u, $u2$, $u0$, ...) are calculated by using the real and imaginary part of the corresponding voltage as follows:

$$u = \sqrt{ur^2 + ui^2} \tag{1}$$

and the voltage difference magnitude is calculated using the real and imaginary voltage difference between the two measurement points:

$$du = \sqrt{(ur_{meas1} - ur_{meas2})^2 + (ui_{meas1} - ui_{meas2})^2} \quad (2)$$

The frequency output based on phase angle derivation is available only for three phase measurement devices (option *No. of Phases = 3*). If the option *Calculate only if signal is connected* is used, the output frequency fe is calculated only if the output signal is connected to another model e.g. via a composite model. Additional possibilities to estimate the frequency are available in the EMT simulation when calculating RMS phasors.

If an element and its controller models are using the A-stable integration algorithm, the connected measurement device should also use this algorithm, thus the flag '*iAstabint*' on the RMS-simulation page should be selected.

The output signals and transformation equations of the *StaVmea* element are documented in the following sections.

2 Load Flow

2.1 Phase quantities

The phase quantities output signals available for the Load Flow calculation are identical to the output signals presented in the RMS simulation. Please refer to Section 3 for more information.

2.2 Symmetrical components

The symmetrical components output signals available for the Load Flow calculation are identical to the output signals presented in the RMS simulation, except that the frequency-related quantities are not available. Please refer to Section 3 for more information.

3 RMS simulation

The output signals available for balanced and unbalanced RMS simulation are described in the following subsections.

3.1 Balanced

From the balanced RMS simulation, the real and imaginary part of the positive-sequence voltage are available. Based on these voltages, also additional voltage quantities can be calculated.

3.1.1 Phase quantities

The available outputs for a measurement at a single terminal are presented in Table 3.1.

Table 3.1: Outputs, Phase quantities for Balanced RMS simulation, single terminal

Name	Unit	Description
$u1r$	p.u.	Positive sequence voltage, real part
$u1i$	p.u.	Positive sequence voltage, imaginary part
$fref$	p.u.	Reference frequency (from the reference machine)
fe	p.u.	Measured frequency

For information on the calculation of fe, please refer to subsection 3.1.2.

The available outputs for a measurement between two terminals are presented in Table 3.2.

Table 3.2: Outputs, Phase quantities for Balanced RMS simulation, between two terminals

Name	Unit	Description
$du1r$	p.u.	Voltage difference, positive sequence, real part
$du1i$	p.u.	Voltage difference, positive sequence, imaginary part

3.1.2 Symmetrical components

The available outputs for a measurement at a single terminal are presented in Table 3.3.

Table 3.3: Outputs, Symmetrical components for Balanced RMS simulation, single terminal

Name	Unit	Description
ur	p.u.	Positive sequence voltage, real part
ui	p.u.	Positive sequence voltage, imaginary part
u	p.u.	Positive sequence voltage, magnitude
$fref$	p.u.	Reference frequency (from the reference machine)
fe	p.u.	Measured frequency

- The frequency fe is calculated by adding the frequency deviation to the frequency of the reference machine. The calculation of the frequency deviation is based on the following equation:

$$\Delta f = \frac{d\,\phi}{dt} \tag{3}$$

Equation 3 is implemented using the following differential equations (two state variables are defined internally $\cos\phi$ and $\sin\phi$) for calculating the frequency output fe:

$$\frac{d\cos\phi}{dt} = \left(\frac{ur}{\sqrt{ur^2+ui^2}} - \cos\phi\right) \Big/ T_{fe} \tag{4}$$

$$\frac{d\sin\phi}{dt} = \left(\frac{ui}{\sqrt{ur^2+ui^2}} - \sin\phi\right) \Big/ T_{fe} \tag{5}$$

where $T_fe = 3/Fnom$ and $Fnom$ is the Nominal Frequency of the network in Hz.

The frequency change (in p.u.) is equal to:

$$dfe = \begin{cases} \left(\frac{d\sin\phi}{dt} \cdot \frac{1}{\cos\phi}\right) \Big/ (2 \cdot \pi \cdot Fnom) & if\ |\cos\phi| > |\sin\phi| \\ -\left(\frac{d\cos\phi}{dt} \cdot \frac{1}{\sin\phi}\right) \Big/ (2 \cdot \pi \cdot Fnom) & if\ |\cos\phi| \leq |\sin\phi| \end{cases} \tag{6}$$

The frequency is then calculated using $fref$ (reference machine frequency in p.u.) as:

$$fe = fref + dfe \tag{7}$$

- For DC systems, only one signal is available ur ($ui = 0$, $fe = 0$):

The available outputs for a measurement between two terminals are presented in Table 3.4.

Table 3.4: Outputs, Symmetrical components for Balanced RMS simulation, between two terminals

Name	Unit	Description
dur	p.u.	Voltage difference, positive sequence, real part
dui	p.u.	Voltage difference, positive sequence, imaginary part
du	p.u.	Voltage difference, positive sequence, magnitude

3.2 Unbalanced

From the unbalanced RMS simulation, the real and imaginary parts of the voltages ($\underline{u}_a$, $\underline{u}_b$, $\underline{u}_c$ and $\underline{u}_n$) are available to the measurement device. Using these available voltages, also additional quantities like the positive-, negative-, and zero-sequence voltages can be calculated.

3.2.1 Phase quantities

The available outputs for a measurement at a single terminal are presented in Table 3.5.

Table 3.5: Outputs, Phase quantities for Unbalanced RMS simulation, single terminal

Name	Unit	Description
ur_A	p.u.	Voltage, phase a, real part
ui_A	p.u.	Voltage, phase a, imaginary part
ur_B	p.u.	Voltage, phase b, real part
ui_B	p.u.	Voltage, phase b, imaginary part
ur_C	p.u.	Voltage, phase c, real part
ui_C	p.u.	Voltage, phase c, imaginary part
$fref$	p.u.	Reference frequency (from the reference machine)
fe	p.u.	Measured frequency

For information on the calculation of fe, please refer to subsection 3.2.2.

The available outputs for a measurement between two terminals are presented in Table 3.6.

Table 3.6: Outputs, Phase quantities for Unbalanced RMS simulation, between two terminals

Name	Unit	Description
dur_A	p.u.	Voltage difference, phase a, real part
dui_A	p.u.	Voltage difference, phase a, imaginary part
dur_B	p.u.	Voltage difference, phase b, real part
dui_B	p.u.	Voltage difference, phase b, imaginary part
dur_C	p.u.	Voltage difference, phase c, real part
dui_C	p.u.	Voltage difference, phase c, imaginary part

3.2.1.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase measurement at a single terminal are presented in Table 3.7.

Table 3.7: Outputs, Phase quantities for Unbalanced RMS simulation, single-phase measurement at a single terminal

Name	Unit	Description
ur	p.u.	Phase voltage, real part
ui	p.u.	Phase voltage, imaginary part

The available outputs for a single-phase measurement (*Number of phases*$= 1$) between two terminals are presented in Table 3.8.

Table 3.8: Outputs, Phase quantities for Unbalanced RMS simulation, single-phase measurement between two terminals

Name	Unit	Description
dur	p.u.	Voltage difference, real part
dui	p.u.	Voltage difference, imaginary part

3.2.2 Symmetrical components

In the case of the three-phase system, the Symmetrical components are the real and imaginary parts and magnitude of the positive-, negative- and zero-sequence voltage that are calculated using the symmetrical components transformation. Similar transformation is being used for the two-phase system (modal components transformation).

The available outputs for a measurement at a single terminal are presented in Table 3.9.

Table 3.9: Outputs, Symmetrical components for Unbalanced RMS simulation, single terminal

Name	Unit	Description
ur	p.u.	Positive sequence voltage, real part
ui	p.u.	Positive sequence voltage, imaginary part
u	p.u.	Positive sequence voltage, magnitude
$u2r$	p.u.	Negative sequence voltage, real part
$u2i$	p.u.	Negative sequence voltage, imaginary part
$u2$	p.u.	Negative sequence voltage, magnitude
$u0r$	p.u.	Zero sequence voltage, real part
$u0i$	p.u.	Zero sequence voltage, imaginary part
$u0$	p.u.	Zero sequence voltage, magnitude
$fref$	p.u.	Reference frequency (from the reference machine)
fe	p.u.	Measured frequency

The output voltages for unbalanced RMS simulation are calculated depending on the phase technology of the terminal as follows:

- 3-phase system (using symmetrical components transformation)

$$ur + j\,ui = \frac{1}{3} \cdot \left(\underline{u}_a + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) \cdot \underline{u}_b + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right) \cdot \underline{u}_c \right) \tag{8}$$

$$u2r + j\,u2i = \frac{1}{3} \cdot \left(\underline{u}_a + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) \cdot \underline{u}_b + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right) \cdot \underline{u}_c \right) \tag{9}$$

$$u0r + j\,u0i = \frac{1}{3} \cdot (\underline{u}_a + \underline{u}_b + \underline{u}_c) \tag{10}$$

- BI-phase system ($180°$):

$$ur + j\,ui = \frac{1}{2} \cdot (\underline{u}_a - \underline{u}_b) \tag{11}$$

$$u2r + j\,u2i = 0 \tag{12}$$

$$u0r + j\,u0i = \frac{1}{2} \cdot (\underline{u}_a + \underline{u}_b) \tag{13}$$

- 2-phase system ($120°$):

$$ur + j\,ui = \frac{1}{\sqrt{3}} \cdot (\underline{u}_a - \underline{u}_b) \tag{14}$$

$$u2r + j\,u2i = 0 \tag{15}$$

$$u0r + j\,u0i = \frac{1}{\sqrt{3}} \cdot (\underline{u}_a + \underline{u}_b) \tag{16}$$

- Single-phase (*Number of phases*$= 3$):

  The voltage is set (transformation is not used) to the complex voltage $\underline{u}$ available from the RMS simulation as follows:

$$ur + j\,ui = \underline{u} \tag{17}$$

- For DC systems, only one signal is available ur ($ui = 0$, $fe = 0$).
- The reference frequency $fref$ is the frequency of the reference machine.
- The frequency fe is calculated using the positive-sequence voltage and the same equations are valid as for the balanced RMS simulation (Equation 3 to Equation 7).

The available outputs for a measurement between two terminals are presented in Table 3.10.

Table 3.10: Outputs, Symmetrical components for Unbalanced RMS simulation, between two terminals

Name	Unit	Description
dur	p.u.	Voltage difference, positive sequence, real part
dui	p.u.	Voltage difference, positive sequence, imaginary part
du	p.u.	Voltage difference, positive sequence, magnitude
$du2r$	p.u.	Voltage difference, negative sequence, real part
$du2i$	p.u.	Voltage difference, negative sequence, imaginary part
$du2$	p.u.	Voltage difference, negative sequence, magnitude
$du0r$	p.u.	Voltage difference, zero sequence, real part
$du0i$	p.u.	Voltage difference, zero sequence, imaginary part
$du0$	p.u.	Voltage difference, zero sequence, magnitude

3.2.2.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase measurement (*Number of phases*$= 1$) at a single terminal are presented in Table 3.11.

Table 3.11: Outputs, Symmetrical components for Unbalanced RMS simulation, single-phase measurement at a single terminal

Name	Unit	Description
ur	p.u.	Voltage, real part
ui	p.u.	Voltage, imaginary part
u	p.u.	Voltage, magnitude

- Single-phase measurement (*Number of phases*$= 1$):

  The voltage is set (transformation is not used) to the complex voltage $\underline{u}$ available from the RMS simulation for the selected measured phase (*a, b, c, a-n, b-n, c-n or n*) as follows:

$$ur + j\, ui = \underline{u} \tag{18}$$

The available outputs for a single-phase measurement (*Number of phases*$= 1$) between two terminals are presented in Table 3.12.

Table 3.12: Outputs, Symmetrical components for Unbalanced RMS simulation, single-phase measurement between two terminals

Name	Unit	Description
dur	p.u.	Voltage difference, real part
dui	p.u.	Voltage difference, imaginary part
du	p.u.	Voltage difference, magnitude

4 EMT Simulation

From the EMT Simulation, the phase voltages (u_a, u_b, u_c and u_n) are available to the measurement device. Using these voltages, additional quantities like the alpha, beta components or RMS voltages can be calculated.

The output signals available for the EMT Simulation are described in the following subsections.

4.1 Instantaneous values (phase quantities)

The available outputs for a instantaneous values measurement at a single terminal are presented in Table 4.1.

Table 4.1: Outputs, Instantaneous values (phase quantities) for EMT Simulation, single terminal

Name	Unit	Description
u_A	p.u.	Voltage A
u_B	p.u.	Voltage B
u_C	p.u.	Voltage C
fe	p.u.	Measured frequency
$phiref$	rad	Reference angle (rotating system)

For information on the calculation of fe and $phiref$, please refer to subsection 4.4.

The available outputs for a instantaneous values measurement between two terminals are presented in Table 4.2.

Table 4.2: Outputs, Instantaneous values (phase quantities) for EMT Simulation, between two terminals

Name	Unit	Description
du_A	p.u.	Voltage difference A
du_B	p.u.	Voltage difference B
du_C	p.u.	Voltage difference C

4.1.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase instantaneous values measurement at a single terminal are presented in Table 4.3.

Table 4.3: Outputs, Instantaneous values (phase quantities) for EMT Simulation, single-phase measurement (*Number of phases*$= 1$) at a single terminal

Name	Unit	Description
u	p.u.	Phase voltage

The available outputs for a single-phase instantaneous values measurement between two terminals are presented in Table 4.4.

Table 4.4: Outputs, Instantaneous values (phase quantities) for EMT Simulation, single-phase measurement between two terminals

Name	Unit	Description
du	p.u.	Voltage difference

4.2 True RMS amplitude values (phase quantities)

The measurement device measuring true RMS values is a clocked device, for which the following input parameters need to be defined:

- *Number of samples per period* (input parameter $samplesPerPeriod$)
- *Calculate (update) outputs every* number of samples (input parameter $samplesForUpdate$)

The resulting sampling frequency in [Hz] with which the calculation buffer is filled at defined intervals is obtained using the nominal frequency $Fnom$ and the input parameter $samplesPerPeriod$ as:

$$samplingFreq = Fnom \cdot samplesPerPeriod \tag{19}$$

The output of the measurement device is calculated at constant intervals after the defined number of samples for updating. The resulting update cycle (parameter $clockPeriod$) in [s] is:

$$clockPeriod = \frac{samplesForUpdate}{samplesPerPeriod \cdot Fnom} \tag{20}$$

The calculation of the true RMS amplitudes from the instantaneous values x_1,x_2, ...$x_{samplesPerPeriod}$ is based on the following equation:

$$RMS = \sqrt{\frac{x_1^2 + x_2^2 + ... + x_{samplesPerPeriod}^2}{samplesPerPeriod}} \tag{21}$$

When the option *Smooth outputs when updated* is used, the outputs are linearly interpolated (smoothed).

The available outputs for a true RMS value measurement at a single terminal are presented in Table 4.5.

Table 4.5: Digital outputs, True RMS amplitude values (phase quantities) for EMT Simulation, single terminal

Name	Unit	Description
u_A	p.u.	RMS voltage A
u_B	p.u.	RMS voltage B
u_C	p.u.	RMS voltage C

4.2.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase instantaneous values measurement at a single terminal are presented in Table 4.6.

Table 4.6: Digital outputs, True RMS amplitude values (phase quantities) for EMT Simulation, single-phase measurement (*Number of phases*$= 1$) at a single terminal

Name	Unit	Description
u	p.u.	RMS voltage

4.3 RMS phasor, symmetrical components and frequency (phase quantities)

The measurement device measuring RMS phasors is a clocked device which calculates a Discrete Fourier Transform (DFT) for a single frequency. There are four different options for estimating the frequency for which the DFT is calculated:

- *Use nominal frequency*

  The DFT is executed for nominal frequency.

- *Phase angle derivative*

  The frequency supplied to the DFT (fe), is estimated based on the phase angle derivative.

  The calculation of the frequency fe is similar to the calculation in the RMS simulation, with the difference that instead the positive-sequence, the dq voltage values are used. The dq quantities of the voltage are obtained as:

$$ud + j\ uq = (ur + j\ ui) \cdot (\cos(phiref) - j\ (\sin(phiref))) \tag{22}$$

  The following differential equations are being used:

$$\frac{d\ \cos\phi}{dt} = \left(\frac{ud}{\sqrt{ud^2 + uq^2}} - \cos\phi \right) \Big/ T_{fe} \tag{23}$$

$$\frac{d\ \sin\phi}{dt} = \left(\frac{uq}{\sqrt{ud^2 + uq^2}} - \sin\phi \right) \Big/ T_{fe} \tag{24}$$

  where T_fe is fixed to $3/Fnom$.

  The frequency change (in p.u.) is equal to:

$$dfe = \sqrt{\frac{\left(\frac{d\ \cos\phi}{dt}\right)^2 + \left(\frac{d\ \sin\phi}{dt}\right)^2}{(\cos\phi)^2 + (\sin\phi)^2}} \Big/ (2 \cdot \pi \cdot Fnom) \tag{25}$$

  The frequency is then calculated as $fe = 1 + dfe$ or as $fe = 1 - dfe$ depending on the angle change.

  Please note that this option works best with balanced, sinusoidal signals and is available only for three-phase systems.

- *Frequency-locked loop*

  The frequency supplied to the DFT (fe), is estimated based on a double second-order generalized integrator frequency-locked loop (DSOGI-FLL) with gain normalisation as presented in [1]. For the single phase measurement, a SOGI-FLL is used for estimating the frequency.

  The following proportional gain parameters need to be defined:
  - *SOGI proportional gain* (input parameter $Ksogi$)
  - *FLL proportional gain* (input parameter $Gfll$)

In some cases, the frequency transients when locking the frequency can be very high, and therefore a rate limiter has been included in order to limit the estimated frequency. If the estimated frequency fe has high transients, a corresponding rate limit in Hz/s (parameter $rateLimit$) can be used in order to provide a "smoother" transient frequency value to the DFT calculation.

Please note that this option works very well also with unbalanced signals. The frequency tracking does not work properly for very strong distorted signals.

- *Fast fourier transformation*

  The frequency supplied to the DFT is obtained by a Fast Fourier Transform (FFT) calculation (separately for each signal) at the clocked interval.

  The following input parameters need to be defined:
  - *Number of samples per period* (input parameter $samplesPerPeriod$)
  - *Number of samples for FFT window* (input parameter $samplesPerWindow$)

  The resulting sampling frequency in Hz, with which the calculation buffer is filled at defined intervals, is obtained using the nominal frequency $Fnom$ and the input parameter $samplesPerPeriod$ as:

$$samplingFreq = Fnom \cdot samplesPerPeriod \tag{26}$$

  The resulting FFT window length in s of the buffer used in the calculation is obtained using the nominal frequency $Fnom$ as:

$$windowLength = \frac{samplesPerWindow}{samplesPerPeriod \cdot Fnom} \tag{27}$$

  Using a moving FFT on the buffer gives us the frequency of the signal with which the DFT is executed.

  Please note that this method has a limited bandwidth and may produce inaccurate results if the frequency deviates from the nominal frequency by more than $\pm 10\%$.

The outputs are updated at constant intervals defined by the parameter *Calculate outputs every* (input parameter $samplesForUpdate$) which defines after how many samples the DFT is calculated (and outputs are updated). The resulting update cycle (parameter $clockPeriod$) in s is:

$$clockPeriod = \frac{samplesForUpdate}{samplesPerPeriod \cdot Fnom} \tag{28}$$

When the option *Smooth outputs when updated* is used, the outputs are linearly interpolated (smoothed).

The available outputs for a FFT/DFT based measurement at a single terminal are presented in Table 4.7.

Table 4.7: Digital outputs, RMS phasor, symmetrical components and frequency (phase quantities) for EMT Simulation, single terminal

Name	Unit	Description
u_A	p.u.	RMS voltage A
u_B	p.u.	RMS voltage B
u_C	p.u.	RMS voltage C
F_A	Hz	Frequency A
F_B	Hz	Frequency B
F_C	Hz	Frequency C
$u1$	p.u.	Positive sequence voltage
$u2$	p.u.	Negative sequence voltage
$u0$	p.u.	Zero sequence voltage
ul_A	p.u.	Line to line RMS voltage A (A-B)
ul_B	p.u.	Line to line RMS voltage B (B-C)
ul_C	p.u.	Line to line RMS voltage C (C-A)

The available calculation parameters for a FFT/DFT based measurement at a single terminal are presented in Table 4.8.

Table 4.8: Calculation quantities, RMS phasor, symmetrical components and frequency (phase quantities) for EMT Simulation, single terminal

Name	Unit	Description
ur:A	p.u.	RMS voltage A, real part
ur:B	p.u.	RMS voltage B, real part
ur:C	p.u.	RMS voltage C, real part
ui:A	p.u.	RMS voltage A, imaginary part
ui:B	p.u.	RMS voltage B, imaginary part
ui:C	p.u.	RMS voltage C, imaginary part

4.3.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase FFT/DFT based measurement at a single terminal are presented in Table 4.9.

Table 4.9: Digital outputs, RMS phasor, symmetrical components and frequency (phase quantities) for EMT Simulation, single-phase measurement (*Number of phases*$= 1$) at a single terminal

Name	Unit	Description
u	p.u.	RMS voltage
F	p.u.	Frequency

The available calculation parameters for a single-phase FFT/DFT based measurement at a single terminal are presented in Table 4.10.

Table 4.10: Calculation quantities, RMS phasor, symmetrical components and frequency (phase quantities) for EMT Simulation, single-phase measurement (*Number of phases*= 1) at a single terminal

Name	Unit	Description
ur	p.u.	RMS voltage, real part
ui	p.u.	RMS voltage, imaginary part

4.4 Alpha, beta and gamma components

The alpha, beta and gamma voltage components are calculated using the Clarke transformation.

The available outputs for a measurement at a single terminal are presented in Table 4.12.

Table 4.11: Outputs, Alpha, beta and gamma components for EMT Simulation, single terminal

Name	Unit	Description
ur	p.u.	Voltage, alpha-component
ui	p.u.	Voltage, beta-component
u	p.u.	Voltage, magnitude (using the alpha and beta component)
$u0$	p.u.	Voltage, gamma-component
fe	p.u.	Measured frequency
$phiref$	rad	Reference angle (rotating system)

The output voltages for the EMT simulation are calculated depending on the phase technology of the terminal as follows:

- 3-phase system:

  The phase voltages are transformed using the $\alpha\beta\gamma$ transformation:

$$ur = u_\alpha = \frac{1}{3} \cdot (2 \cdot u_a - u_b - u_c) \tag{29}$$

$$ui = u_\beta = \frac{1}{3} \cdot (\sqrt{3} \cdot u_b - \sqrt{3} \cdot u_c) \tag{30}$$

$$u0 = u_\gamma = \frac{1}{3} \cdot (u_a + u_b + u_c) \tag{31}$$

- BI-phase system ($180°$):

$$ur = \frac{1}{2} \cdot (u_a - u_b) \tag{32}$$

$$ui = 0 \tag{33}$$

$$u0 = \frac{1}{2} \cdot (u_a + u_b) \tag{34}$$

- 2-phase system ($120°$):

$$ur = \frac{1}{\sqrt{3}} \cdot (u_a - u_b) \tag{35}$$

$$ui = 0 \tag{36}$$

$$u0 = \frac{1}{\sqrt{3}} \cdot (u_a + u_b) \tag{37}$$

- Single-phase (*Number of phases*$= 3$):

  The voltage is set (transformation is not used) to the voltage u available from the EMT Simulation as follows:

$$ur = u \tag{38}$$
$$ui = 0 \tag{39}$$
$$u0 = 0 \tag{40}$$

- For DC systems, only one signal is available ur ($ui = 0$, $u0 = 0$, $fe = 0$).
- The angle $phiref$ in rad is used when calculating the frequency (artificially created rotating system). Its time derivative is calculated using the nominal frequency of the network $Fnom$ in Hz as follows:

$$\frac{d\,phiref}{dt} = 2 \cdot \pi \cdot Fnom \tag{41}$$

- The calculation of the frequency fe is similar to the calculation in the RMS simulation, with the difference that instead the positive-sequence, the dq voltage values are used. The dq quantities of the voltage are obtained as:

$$ud + j\,uq = (ur + j\,ui) \cdot (\cos(phiref) - j\,(\sin(phiref))) \tag{42}$$

  The following differential equations are being used:

$$\frac{d\,\cos\phi}{dt} = \left(\frac{ud}{\sqrt{ud^2 + uq^2}} - \cos\phi\right) \Big/ T_{fe} \tag{43}$$
$$\frac{d\,\sin\phi}{dt} = \left(\frac{uq}{\sqrt{ud^2 + uq^2}} - \sin\phi\right) \Big/ T_{fe} \tag{44}$$

  where T_fe is fixed to $3/Fnom$.

  The frequency change (in p.u.) is equal to:

$$dfe = \sqrt{\frac{\left(\frac{d\,\cos\phi}{dt}\right)^2 + \left(\frac{d\,\sin\phi}{dt}\right)^2}{(\cos\phi)^2 + (\sin\phi)^2}} \Big/ (2 \cdot \pi \cdot Fnom) \tag{45}$$

  The frequency is then calculated as $fe = 1 + dfe$ or as $fe = 1 - dfe$ depending on the angle change.

The available outputs for a measurement between two terminals are presented in Table 4.12.

Table 4.12: Outputs, Alpha, beta and gamma components for EMT Simulation, between two terminals

Name	Unit	Description
dur	p.u.	Voltage difference, alpha-componentt
dui	p.u.	Voltage difference, beta-component
du	p.u.	Voltage difference, magnitude (using the alpha and beta component)
$du0$	p.u.	Voltage difference, gamma-component

4.4.0.1 Single-phase measurement (*Number of phases*$= 1$)

The available outputs for a single-phase measurement (*Number of phases*$= 1$) at a single terminal are presented in Table 4.13.

Table 4.13: Outputs, Alpha, beta and gamma components for EMT Simulation, single-phase measurement at a single terminal

Name	Unit	Description
ur	p.u.	Voltage
ui	p.u.	Voltage (set to 0)
u	p.u.	Voltage, magnitude

- Single-phase measurement (*Number of phases*$= 1$):

  The voltage is set (transformation is not used) to the voltage u_{phase} available from the EMT Simulation for the selected measured phase (*a, b, c, a-n, b-n, c-n or n*) as follows:

$$ur = u_{phase} \tag{46}$$

$$ui = 0 \tag{47}$$

The available outputs for a single-phase measurement (*Number of phases*$= 1$) between two terminals are presented in Table 4.14.

Table 4.14: Outputs, Alpha, beta and gamma components for EMT Simulation, single-phase measurement between two terminals

Name	Unit	Description
du	p.u.	Voltage difference, magnitude

A References

[1] R. Teodorescu, M. Liserre, and P. Rodriguez. *Grid Converters for Photovoltaic and Wind Power Systems*. Wiley, 2011.

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